

## Smart Sensors for Aircraft

The StrainWISE project, initiated by Airbus as part of the 'Clean Sky' programme, resulted in the development of a high-performance and ultra-low-power wireless sensor network platform that is able to monitor aircrafts' critical structures.

The proposed solution, which won the top prize in the Research category at the Mechatronics Awards 2013, allows the installation of a large number of sensors, without any restrictions on their location, hence providing additional or more precise data. The new information, provided without any additional wires, allows for an improved control of the aircraft and presents major advantages compared to traditional solutions, notably in terms of fuel consumption and maintenance costs.

Each sensor is equipped with a radio transceiver and an independent power supply. Energy consumption, both during flight and while the aircraft is on the ground, has been reduced to a minimum by a combination of energy-efficient electronic components and by the optimisation of the communication protocol.

The sensor is powered by energy harvesting from a thermoelectric unit, removing the need to change batteries throughout the sensors' 35-year life. Thanks to this unlimited autonomy, a wide array of sensors can now be placed very close to the structures to be monitored, at places that would be very difficult to access for maintenance, such as in the fuselage, the wings, and the stabiliser.

Each sensing node allows the measurements to be timed with an accuracy of less than a millisecond, for sampling frequencies ranging from 120 to 500Hz. The information is communicated to the plane computers by radio. They do not interfere with the operation of the pre-existing devices, as the system successfully passed the qualification tests related to radio interferences.

A demonstrator has been delivered to Airbus and is currently undergoing tests. The partners are pursuing their collaboration as part of a new Clean Sky project called FLITE-WISE, the objective of which is to create a new platform of high-speed autonomous sensors intended for use on the outside of the aircraft, both on fixed and moving parts.

In node of the sensor network, the metal plate is positioned against the interior of the aircraft's cabin. The red tank (thermal insulator) contains water, whose conversion to solid state is delayed when the aircraft climbs in order to create a temperature differential at the extremities of the TEG cell, which is converted into electrical current. The green circuit board controls energy flow and recharging of the battery. The blue board contains the sensor interface, the radio frequency circuit, the antennas, and the microcontroller—the heart of the system's intelligence.



*A node in the sensor network*

flight phases except the taxing. Landing on runway and controlling the aircraft on rollout is Cat-3B landing used in majority of the airports. Landing, rollout, and taxi control is Cat-3C. Cat-3C autopilots will be used in the future.

Autopilots use computer software to control the aircraft. The software reads the aircraft's current position and controls flight control systems to guide the aircraft. In addition to flight control many autopilots incorporate thrust control capabilities that can control throttle of engines. In short, the autopilots integrate both the fly-by-wire of flight control and power-by-wire of engine control to optimise the flight path of the aircraft from take-off to landing and reduce the workload of the pilot drastically.

The following are the electrical actuation systems in most of the aircrafts:

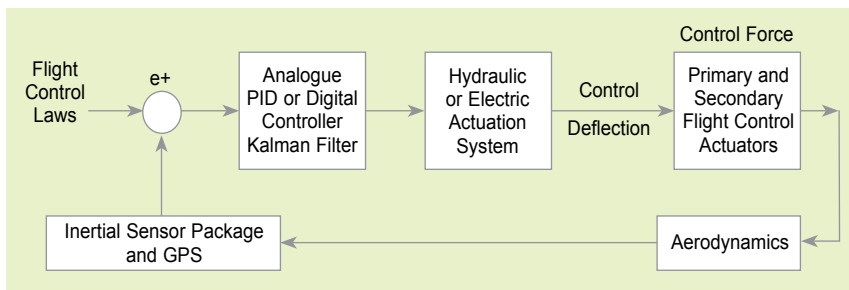
1. Landing gear actuation
2. Primary flight controls
3. Secondary flight controls
4. Thrust reverser actuation
5. Various types of door actuators and utilities

## Conclusion

Like the human nervous system, aircraft sensors are located at various parts of the aircraft to monitor, diagnose, and eliminate the faulty sensors during flight. These wireless systems are lightweight, easy to install, and allow for scalable sensing at a fraction of their weight and size. The sensors are powered by energy harvesting from a thermoelectric unit, removing the need to change batteries throughout the sensors' 35-year life.

A bank of digital Kalman filters eliminates the faulty sensors, specially on the engine during flight, so that the aircraft and engine performance is not adversely affected.

Recent advances in the BLDC motor technology are now successfully adapted for electric aircraft propulsion as well as flight control and autopilot actuation systems. **EFY**



*Autopilot block diagram*

For all the low and medium torque applications electrical actuators are replacing electro-hydraulic actuators in aerospace applications. These are compact in size and are maintenance-free. They are driven by a brushless DC motor as the prime mover.

The BLDC motor uses high-energy magnets like neodymium-iron-boron or samarium-cobalt magnets.

A modern autopilot divides a flight into a taxi, take-off, ascent, level, descent, approach, landing, and taxi phases. Autopilots automate all these

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